Team Selection Test Semester 1

Paly Math Club

2025-26

This test is the team-selection test for **Berkeley Math Tournament**, **Princeton Mathematics Competition**, **Caltech Math Meet**, and **Stanford Math Tournament**.

There will likely be a second team selection test next semester when choosing CMM and SMT, which will be considered more in team selection.

The following are brief information of the competitions:

BMT	PUMaC	CMM	SMT
11/8/2025	11/22/2025	01/24/2026	April 2026
Cost: $0	Cost: $1000	Cost: $250	Cost: $0
Location: UC Berkeley	Location: Princeton	Location: Caltech	Location: Stanford

Your decision of BMT and PUMaC is BINDING. Withdrawing on short notice may prohibit you from participating in future tournaments this year.

Additional Note: PUMaC happens during the weekend before Thanksgiving.

Please circle the competitions you are planning to attend:

1) BMT 2) PUMaC 3) CMM 4) SMT

Instructions: This is a **60** minute test with **15** questions. Each question is given 1 point. No point deduction for incorrect answers, no partial credit except the estimation problem.

Name: _____ Grade: _____

1 Algebra

1. What is the value of $(3+2\sqrt{2})+\left(\frac{1}{3-2\sqrt{2}}\right)+(3-2\sqrt{2})+\left(\frac{1}{3+2\sqrt{2}}\right)$?

2. Let $f(z)=\dfrac{(z+\overline{z})^2}{\overline{z}}-(z+\overline{z})$ for all nonzero complex numbers z. (*Note: if $z=a+bi$, then $\overline{z}=a-bi$.*) If $f(z)=1+i\sqrt{3}$, what is the sum of all possible values of z?

3. A sequence a_n is defined in the following way for all natural n:

$$a_1=\frac{1}{2},\ a_{n+1}=\frac{n+1}{n^2+2n}a_n$$

Find

$$\sum_{n=1}^{\infty}a_n.$$

4. Let $\omega=e^{2\pi i/7}$. Find

$$\sum_{i=1}^{7}\sum_{j=1}^{7}\frac{1+\omega^i}{1+\omega^j}.$$

2 Geometry

1. Sky He decides to construct a hexagonal prism with the side lengths of every side being equal to 3. What is the volume?

2. Suppose points A, B, C, D, E, and F lie on a line such that

$$AB=1,\quad BC=4,\quad CD=9,\quad DE=16,\quad EF=25,\quad \text{and}\quad FA=23.$$

What is CF?

3. Let a and b be real numbers such that $z_1=1010+2021i$, $z_2=2020+1011i$, and $z_3=az_1+bz_2$ are the vertices of an equilateral triangle in the complex plane. What is the sum of all possible values of $a+b$?

4. In $\triangle ABC$, $AB=8$, $AC=10$ and $BC=12$. Suppose O is the circumcenter of $\triangle ABC$ and M is the midpoint of $\overline{AB}$. Let P be the point such that P is in the interior of $\triangle AOB$, $PA=PB$ and $\frac{PM}{OM}=\frac{1}{4}$. Find $\sin(\angle PAB)$.

3 Combinatorics

1. How many ordered triples of integers (a,b,c) satisfy

$$2\le|a|+|b|+|c|\le 5\,?$$

2. There is a 7 by 7 grid of dots (49 dots total). The top left 2 by 2 square of dots (4 dots) is removed, and the bottom right 3 by 3 square of dots (9 dots) is removed. How many paths are there from the bottom left dot to the top right dot assuming that you can only go up or right and you can only turn at a dot?

3. The constant term in the expansion of $(x+\frac{p}{x})^n$ is equal to 160, where p is a prime number. Find $n\cdot p$, the product of n and p.

4 Number Theory

1. Aiden Yuan decides to go to a confectionery store to satisfy his big back tendencies. His particular store of choice only sells candy in packs of 6, 7, or 41. What is the largest amount of candy that he cannot buy?

2. What is
$$\sum_{n=1}^{105} \gcd(n, 105)?$$

3. How many ordered triples a, b, c are there such that $0 \leq a, b, c < 11$ and $a^5 + b^5 \equiv c^5 \pmod{11}$?

5 Estimation

We are told that 843301 is a prime number. For a prime p, the *primorial* $p\#$ means the product of all primes up to and including p. For example,
$$13\# = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13.$$

How many base-10 digits does the number 843301# have (i.e., how many digits appear when 843301# is written in decimal)?[1]

[1] Your score is scaled to the interval $[0, 1]$ by
$$S = \max\left\{ \frac{1}{20} \left\lfloor 60\left(\frac{1}{3} - \left|\ln(A/d)\right|\right)\right\rfloor,\ 0\right\},$$
where A is the value you submit and d is the correct answer.